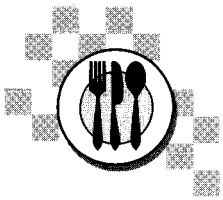
CHAPTER 14



Fruit and Vegetable Processing and Product Sanitation

About This Chapter

In this chapter you will learn:

1. How to reduce contamination of fruit and vegetable products from raw products, soil, air, water, and pests
2. How good housekeeping and waste-disposal practices can help keep fruit and vegetable products wholesome
3. How the design of fruit- and vegetable-processing plants can affect hygiene
4. Methods for cleaning and sanitizing in fruit- and vegetable-processing plants
5. How sanitarians evaluate hygiene in fruit- and vegetable-processing plants

Introduction

Like other food operations, facilities that process fruits and vegetables should have a sanitation program that uses correct cleaning compounds, sanitizers, and cleaning procedures. Management must oversee and evaluate the sanitation program. The ultimate goal is to produce food products that are sanitary and wholesome.

Reducing Contamination

Effective sanitation prevents microorganisms that cause spoilage and food poisoning from getting into fruits and vegetables during production, processing, storage, and distribution. It is important to remember that raw fruits and vegetables can have microorganisms on the surface or inside when they arrive at the plant. These microorganisms can contaminate the processing plant.

Federal laws require that processed foods be free of pathogenic microorganisms if they are shipped between states. For commercially canned foods, the normal sterilization process destroys all pathogenic bacteria in the container. Washing and peeling of fruits and vegetables also help remove microorganisms. Therefore, properly canned and frozen fruits and vegetables should be wholesome.

Contamination from Soil

The soil in which vegetables grow contains heat-resistant bacteria. If processors do not wash vegetables thoroughly before they are canned, these bacteria can cause a sour taste and aroma and other types of spoilage. The number and types of microbes are affected by wind, humidity, sunlight, temperature, domestic and wild animals, irrigation water, bird droppings, harvesting equipment, and workers. Many pathogens reach fruits and vegetables during irrigation shortly before harvesting and before the sun dehydrates and destroys the pathogens. Others reach them during harvesting and shipping.

Contamination from Air

Contaminated air transports microorganisms (including pathogens) and pollutants. Air filters can prevent unclean air from entering the food-processing facility.

Contamination by Pests

Some pests invade fruits and vegetables while they are forming on the tree or vine. Pests spread viruses, spoilage bacteria, and pathogens, and also cause physical damage to the fruit or vegetable. The pests are often inactive while the fruit or vegetable is forming because their skins keep the pests from getting inside, and there is very little moisture (low A_w) on the surface. As the fruit or vegetable matures, the structure changes and the pests can grow. When the protective skins of fruits and vegetables are broken by a bruise, mechanical injury, or attack by insects, microorganisms can get inside.

The pollinating fig wasp is a pest that carries microbes that live and grow in the fig while it ripens. Some microbes do not cause spoilage themselves, but they attract other organisms, such as fruit flies, which carry spoilage yeasts and bacteria.

Pests also enter fruits and vegetables during transport. Produce is especially vulnerable at this time if it has cuts or bruises from harvesting.

Processors should not use recirculated water for washing fruits and vegetables because microorganisms build up in the washwater. Chlorinating the washwater does not help because bacterial spores are resistant to chlorine.

Contamination During Processing

Traditional fruit and vegetable canning involved pouring food into containers (metal, glass, or plastic), sealing, and heat treatment. The heat treatment is called *terminal sterilization.* The time and temperature are carefully planned to kill *Clostridium botu-*

linum spores and most heat-resistant spores of other spoilage organisms. Canned foods have what is called "commercial sterility" (i.e., not quite sterile but enough bacteria and spores are destroyed for the food to be safe). In aseptic packaging or aseptic canning, the food and containers are commercially sterilized separately. The food is cooled, and the containers are filled and sealed in aseptic conditions.

In terminal sterilization, the sealed containers are heated (kill step). Because the containers are well made and durable, conventional canning is a very safe way to process food. This technology can also use HACCP to ensure safety.

Aseptic packaging is a relatively new technology. Therefore, it is important to find the best test methods to make sure that the package stays intact and sterile during distribution.

SANITATION MANAGEMENT

As in other food plants, managers are legally and morally responsible for providing consumers with wholesome products. An effective sanitation program keeps the processing environment clean.

Housekeeping

Housekeeping means being organized and tidy. Supervisors should carefully arrange supplies, materials, and clothing to keep the operation tidy, thus less likely to be contaminated and easier to clean.

The sanitarian should be responsible for housekeeping, but all production, maintenance, and sanitation workers need to work together. Employees must keep trash containers, tools, supplies, and personal belongings in their proper places. Trash is more likely to be thrown away immediately if trash containers are in convenient places.

Insects, rodents, and birds can contaminate foods. When sanitation is good, pests cannot find food and shelter in the plant. Use of air and mesh screens; filling in holes, cracks, and crevices; and other hygienic designs help keep pests out of the plant. Regular pest inspections are also important. (Pest control is discussed in Chap. 9.)

Waste Disposal

Waste is easier to handle and by-products are easier to salvage when solid and liquid wastes are separated. Waste disposal is discussed in Chapter 7. Some food-processing plants use waste by-products. The citrus juice industry is very efficient at salvaging waste and keeping waste-disposal costs down. More than 99% of the raw fruit is made into juices, concentrates, or dried cattle feed.

Water Supply

A plentiful supply of clean water is needed for a wholesome product and a clean plant. Water is not just a cleaning medium; it is also used to heat and cool products, and as an ingredient.

Staff should check levels of bacteria and organic or inorganic impurities in the water every day. The number of bacteria determines whether the water is safe to use as a food ingredient or to clean any surface that could touch or contaminate the food. Organic and inorganic impurities affect how well the water will wash the food, surfaces, and equipment.

PLANT CONSTRUCTION

Even a well-designed processing plant does not stop microbes from reaching the food unless each area and each piece of equipment are easy to clean. Proper instructions should be posted for each cleaning operation. If a processing plant is being newly built, expanded, or renovated, all professional staff (mechanical engineers, industrial engineers, food chemists, microbiologists, sanitarians, and operations personnel) should study the functional layouts, mechanical layouts, plumbing layouts, and equipment and construction specifications. Input from each of these people helps make sure that nothing, including sanitation, is overlooked.

Today, new and expanded fruit- and vegetable-processing plants must use hygienic designs because they process such a high volume. High-volume plants are more mechanized. Mechanized startup, shutdown, cleaning, and sanitizing mean less chance for human error but more chance of missing cleaning errors. Highly mechanized plants use more cleaning-in-place (CIP) systems, and less manual cleaning, and fewer visual inspections. However, CIP equipment is still rare in fruit- and vegetable-processing plants, except those that manufacture juices.

High-volume processing plants have longer production periods than lower-volume plants. More microbes build up in the plant because they have a longer time to accumulate and more food goes through the system. Saturation devices can (1) sense when microbes reach an unsafe level, (2) stop production, and (3) trigger an automatic cleaning procedure. These devices should probably be set so that they work only if microbe levels are very high, for example, 150% of normal levels.

Sanitary design features keep downtime for cleaning and sterilizing to a minimum. To be cost effective, equipment and facilities need to operate as much as possible and produce as little sewage as possible.

Processors can now use mechanized and automatic systems to clean equipment that used to have to be cleaned by hand. Before CIP was developed, machines and storage equipment were taken apart at the end of every production day and hand cleaned. CIP used to be controlled using a control panel with push buttons. Now panels have computer-controlled timers that automatically start and stop cleaning, rinsing, and sanitizing steps. (CIP is discussed in Chap. 7.)

Hygienic plant designs have no crevices (narrow and deep cracks or openings) or pockets (large cracks and openings) in the buildings and equipment. Crevices are harder to clean than pockets because they are harder to reach.

Principles of Hygienic Design

Newly constructed or remodeled fruit- or vegetable-processing plants should meet at least the following hygienic design standards:

- Equipment should be designed so that workers can take apart all surfaces that touch the food for manual cleaning or CIP.
- Outside surfaces should not allow soil, pests, and microorganisms to collect on the equipment or in the production area (walls, floors, ceilings, hanging supports, etc.).
- The equipment should protect food from contamination.
- All surfaces that touch food should be inert, so that they do not react with the food and are not absorbed by the food.
- All surfaces that touch the food should be smooth and nonporous, so that food particles, insect eggs, and microorganisms cannot collect in tiny cracks or scratches on the surface.
- The inside of the equipment should have as few crevices and pockets as possible.

The inside and outside of the plant should be sanitary. The following should be avoided so that debris cannot collect:

- Ledges and dirt traps
- Bolts, screws, and rivets that stick out
- Recessed corners, uneven surfaces, and hollows
- Sharp edges
- Unfilled edges

The plant should have double-door construction, heavy-duty weather strips, and self-closing door mechanisms to prevent pests from entering the plant.

It is important to make sure that the facility is designed and built properly, otherwise it may need expensive renovations. Also, the plant must be built carefully to make sure that areas and equipment are not contaminated during construction. The layout must be flexible, because the processing system may change and use new technology. The following points can help reduce contamination:

- The plant should have adequate storage space for raw materials and supplies. The area should be big enough to inspect raw materials before they are stored to make sure they are not carrying pests or dirt. Contaminated supplies should be separated from the rest and cleaned so that contamination does not spread. Cleaning and maintenance materials can taint raw materials if they share the same storage area.
- Finished food products should be stored separately from raw materials. If there is not enough space, the production area may be used to store the finished product. However, this can cause cross-contamination of raw materials while they are being processed.
- Open food production areas should not be blocked or cluttered, so that staff can clean and maintain equipment easily. Staff injuries and equipment damage are more common in messy work areas.
- Waste-removal routes should be short and direct, so that staff do not carry waste through open production areas. This is important because waste collection equipment is often unsanitary.

- Returned goods are often infested or partially decomposed. They must be stored away from raw material and production areas.
- The area around the building should be designed to keep pests out. Waste should be collected, treated, and incinerated away from the plant and away from air intake vents. The outside area should be easy to clean, well drained, and free of weeds, discarded equipment, and surplus supplies.
- Employees must have excellent personal hygiene (see Chap. 4).

CLEANING PROCESSING PLANTS

Cleaning equipment needs an efficient layout to save cleaning time. If possible, the cleaning layout should be designed when the facility is built, expanded, or renovated. It is much easier to install cleaning equipment at the same time as the processing equipment than to add it later. The soil found in fruit- and vegetable-processing plants is easy to remove using portable cleaning systems in small plants and a combination of CIP and centralized foam cleaning in large plants.

Hot-Water Wash

Water carries cleaning compounds and suspended soil. It removes sugars, other carbohydrates, and other compounds that dissolve in water. A hot-water (60–80°C, 140–176°F) wash requires very little cleaning equipment. But it uses a lot of labor and energy and causes condensation on equipment and surfaces. This cleaning technique does not remove heavy soil deposits very well.

High-Pressure, Low-Volume Cleaning

High-pressure spray cleaning works well for removing heavy soils in the fruit- and vegetable-processing industry. This method can clean difficult-to-reach areas with less labor, and the cleaning compounds work better at temperatures below 60°C (140°F). Water that is hotter than this can bake the soil onto the surface and speed up growth of microbes.

A high-pressure, low-volume system uses portable hydraulic units that can be easily moved throughout the plant. Portable equipment works well for conveyors and processing equipment where soaking is impractical and manual cleaning is difficult and time consuming. Centralized high-pressure, low-volume equipment uses the same principle as the portable unit but provides a higher output per man-hour of labor. Most suppliers of these systems can give customers technical assistance to make sure the equipment and cleaning compounds work properly.

Foam Cleaning

Portable foam cleaning is used a lot in fruit- and vegetable-processing plants because the foam is quick and easy to apply to ceilings, walls, piping, belts, and storage con-

tainers. The equipment is about the same size and costs about the same as portable high-pressure units.

Centralized foam cleaning uses the same technique as portable foam equipment and the equipment is installed at convenient places in the plant. The cleaning compound mixes with water and air automatically to form a foam.

Gel Cleaning

With this method, the cleaning compound is used as a gel rather than as a spray or foam. Gel works well for cleaning, canning, and packaging equipment because it clings to the surface.

Slurry Cleaning

This method is the same as foam cleaning, except that less air is mixed with the cleaning compound. A slurry is more liquid than foam and penetrates uneven surfaces in a canning plant better, but it does not cling as well as foam.

Combination Centralized High-Pressure and Foam Cleaning

This system is the same as centralized high pressure, except that the equipment can also apply foam. This method is more flexible because foam can be used for large surface areas, and high pressure can be used for belts, stainless-steel conveyors, and hard-to-reach areas in a canning plant.

Cleaning in Place (CIP)

With this closed system, a recirculating cleaning solution is sprayed into the equipment through nozzles and automatically cleans, rinses, and sanitizes equipment. CIP equipment is expensive and does not work well on heavy soil. But CIP cleaning is used in vacuum chambers, pumping and circulation lines, and large storage tanks. High-volume operations use CIP cleaning because it saves so much labor, which pays for the equipment more quickly. See Chapter 7 for more information about cleaning equipment.

CLEANERS AND SANITIZERS

Soil that is not removed during cleaning is contaminated with microorganisms. Soil can protect microorganisms so that they do not come in contact with chemical sanitizers. Also, soil dilutes chemical sanitizing solutions.

If the soil is light and the temperature of the cleaning solution is below 60°C (140°F), a one-step combined chemical sanitizer and cleaning compound may be used. Combination cleaners (detergent-sanitizers) are often used for manual cleaning in smaller operations with a solution temperature below 60°C (140°F). If the clean-

ing solution is hotter than 80°C (176°F), the solution will destroy most pathogenic and spoilage microorganisms without using a chemical sanitizer. The best chemical sanitizers for fruit- and vegetable-processing plants are halogen compounds, quaternary ammonium compounds, and phenolic compounds.

Halogen Compounds

Chlorine and chlorine compounds are the best halogen sanitizers for food-processing equipment and containers and for water supplies. Calcium hypochlorite and sodium hypochlorite are often used in fruit- and vegetable-processing plants. Plain chlorine is cheaper, but calcium hypochlorite and sodium hypochlorite are easier to use in dilute solution. Hypochlorite solutions are sensitive to temperature changes, residual organic soil, and pH (acidity). These compounds work quickly and are cheaper than other halogens, but they tend to be more corrosive and irritating to the skin. Chapter 6 has more information about chlorine and iodine sanitizers.

Quaternary Ammonium Compounds

Quaternary ammonium compounds ("quats") work well against most bacteria and molds. Quats are stable as a dry powder, a concentrated paste, or in solution at room temperature. They are not affected by heat, dissolve in water, have no color or smell, do not corrode metals, and do not irritate skin. These compounds work better than other sanitizers when there are soil residues. They work best against microbes when the pH is slightly acid or is alkaline (pH ≥6.0). The quats do not work well against bacteria when they are combined with cleaning compounds or in hard water.

Phenolic Compounds

These compounds are often used in antifungal paints and antifungal protective coatings, rather than as sanitizers applied after cleaning. Phenolic compounds have little use in fruit and vegetable plants because they do not dissolve well in water.

CLEANING PROCEDURES

As with other types of food plant, one set of procedures will not work for every fruit- and vegetable-processing plant. Procedures will vary with the design, size, age, and condition of the plant. The guidelines that follow should be adapted for each operation.

Making Cleaning Easier

Cleaning is easier if employees take steps to reduce the amount of soil and clean or rinse equipment quickly after use.

1. Control heating of equipment so that soil does not burn on.
2. Rinse and wash equipment immediately after use so that soil does not dry on.

3. Replace faulty gaskets and seals so that they do not leak and splatter.
4. Handle food products and ingredients carefully so that they do not spill.
5. Keep all areas tidy.
6. If equipment breaks down, rinse equipment and cool to 35°C (95°F) to slow down growth of microbes.
7. During brief shutdowns, keep washers, dewatering screens, blanchers, and similar equipment running and cooled to 35°C (95°F) or below.

Preparing to Clean

1. Remove large debris from the area.
2. Take equipment apart as much as possible.
3. Cover electrical connections with plastic film.
4. Disconnect lines or open cutouts so that debris is not washed onto other equipment that has been cleaned.
5. Remove large chunks of waste from equipment using an air hose, broom, shovel, or other appropriate tool.

Processing Areas

Frequency. Daily.

Procedure

1. Prerinse all soiled surfaces with water at 55°C (131°F) to move dirt down from the ceilings and walls to the floor drains. Do not hose motors, outlets, and electrical cables directly.
2. Use portable or centralized foam cleaning equipment and a strongly acid cleaner. A centralized system works best for large plants; portable equipment is more practical for smaller plants. If the equipment is made of a metal other than stainless steel, a heavy-duty alkaline cleaning compound should be used, rather than an acid cleaner. If necessary, hand-brush soil deposits that stick after foam cleaning. The cleaning compound should reach all framework, table undersides, and other difficult-to-reach areas. The cleaning compound should soak for 10 to 20 minutes.
3. Rinse surfaces within 20 minutes after applying the cleaner to remove the residues. The area should be rinsed from top to bottom using water at 50–55°C (122–131°F).
4. Inspect all surfaces thoroughly, and do touch-up cleaning, if necessary.
5. Use centralized or portable sanitizing equipment and a chlorine compound sanitizer. (The concentration should be 100 ppm of chlorine.) Use the same method to sanitize water pipes used to recirculate washwater and to pump brines, syrups, and vegetables such as peas and corn. Frequently drain, clean, and sanitize water storage tanks so that microbes do not build up.
6. Thoroughly backwash and sanitize water filters and water softeners.

7. Remove scale from the surfaces of pipes and equipment, because scale can collect microorganisms.
8. Remove, clean, and replace drain covers.
9. Apply a white edible oil only to surfaces that may rust or corrode. Do not use oil anywhere else because it can collect microorganisms.
10. Require that maintenance workers carry a sanitizer and use it where they have worked.

Large processing plants can use a CIP system for cleaning piping, large storage tanks, and cookers. The CIP system can be used instead of steps 1, 2, 3, and 5.

Packaged Storage Areas

Frequency. At least once a week where processed products are stored, and more often in a high-volume operation; daily in areas where raw products are stored.

Procedure

1. Pick up large debris and place in trashcans.
2. Sweep and/or scrub the area. Use a mechanical sweeper or scrubber, if possible. Use the correct cleaning compounds for the mechanical sweeper or scrubber, and follow the vendor's directions.
3. Use a portable or centralized foam or slurry cleaning system with water at 50°C (122°F) to clean heavily soiled areas. Rinse as described for the processing areas.
4. Remove, clean, and replace drain covers.
5. Replace hoses and other equipment.
6. Wash and sanitize vegetable boxes each time they are emptied. Replace wooden husker and cutter bins with metal containers.

EVALUATION

The sanitarian must evaluate the sanitation program to see if all areas and equipment are clean and sanitary. Written records are important to show that the facility is meeting its sanitation goals.

Laboratory Tests

The sanitarian must know the type, characteristics, and sources of microorganisms found in the plant so that he or she knows which laboratory tests to use. Laboratory tests can help monitor how well the sanitation program is working. The sanitarian should try to reduce the total number of microorganisms on clean equipment and in processed food products. However, the total number of microbes is not always related to the number of spoilage or pathogenic microorganisms. The sanitarian needs to identify coliforms because they indicate contamination, and identify thermophiles and certain mesophiles because they may cause spoilage. Large numbers of spore-

forming bacteria can shorten the shelf life of the food product and may cause foodborne illness.

Spot laboratory checks for microbial load can be used when visual inspections suggest a problem. The sanitarian should take samples from products and equipment at various stages during processing to identify trouble spots. Regular laboratory tests encourage workers to think about and use good sanitary practices.

SUMMARY

- A good sanitation program for fruit- and vegetable-processing facilities includes:

  Hygienic design of facilities and equipment

  Training of sanitation personnel

  Use of the right cleaning compounds and sanitizers

  Use of good cleaning procedures

  Good administration of the sanitation program

  Evaluation of the program using visual inspection and laboratory tests

- Contamination of raw materials, water, air, and supplies should be kept to a minimum.
- Hygienic design makes the equipment and facility easier to clean.
- Portable or centralized high-pressure or foam cleaning systems and CIP systems can reduce cleaning labor.
- Acid cleaning compounds work well in many facilities, if equipment and surfaces are made of durable material. Chlorine compound sanitizers work best and are most economical. Phenolic compounds can be added to paints and other protective coatings.
- The sanitarian should use visual inspection and laboratory tests to evaluate the sanitation program.

BIBLIOGRAPHY

Jowitt, R. 1980. *Hygienic Design and Operation of Food Plant.* AVI Publishing Co., Westport, Conn.

Lopez, A. 1987. *A Complete Course in Canning (Book I),* 12th ed. Canning Trade, Baltimore.

Marriott, N. G. 1994. *Principles of Food Sanitation,* 3d ed. Chapman & Hall, New York.

STUDY QUESTIONS

1. How are microorganisms destroyed during canning?
2. What are the three main sources of contamination for processed fruits and vegetables?

3. Why is it important to keep production areas tidy?
4. Why is it important to keep the outside of the building neat and clean?
5. What types of fruit- and vegetable-processing plants are likely to use CIP?
6. What is the difference between traditional canning and aseptic canning?
7. Why does the sanitarian need to know which microorganisms are likely to be present?

TO FIND OUT MORE ABOUT FRUIT AND VEGETABLE PROCESSING AND SANITATION

1. Contact the Produce Marketing Association, 700 Barksdale Plaza, Newark, DE 19711, (302) 738-7100. Ask for information about food safety and sanitation.
2. Contact the National Grocers Association, 1825 Samuel Morse Drive, Reston, VA 22090-5317, (703) 437-5300. Ask for information about food safety and sanitation.



RAW MATERIALS CAN BE CONTAMINATED!

Farmers need to be careful that food is not contaminated at the farm after it is harvested. *Cryptosporidium* is an intestinal parasite that causes diarrhea. It can be transmitted by contaminated water (waterborne) or from person to person (for example, in child care or hospital settings, where people who change diapers or help people use the toilet also serve food). But in October 1993, fresh-pressed apple cider from contaminated apples caused a rare case of foodborne cryptosporidiosis.

Two elementary schools and a high school in a rural farming community in central Maine held a 1-day agricultural fair. Local farmers provided agricultural demonstrations, a petting zoo of farm animals, a hay ride, a cider-pressing demonstration, and light refreshments. Eight days after the fair, principals of both elementary schools noticed that many children who had attended the fair were absent from classes.

Investigators collected evidence that showed this was a foodborne outbreak of cryptosporidiosis.

- Of the 759 people who went to the fair, 611 completed surveys.
- People who had diarrhea for at least 3 days totaled 160.
- Of those people who were ill, 56 provided stool samples for testing; 50 samples contained *Cryptosporidium.*
- Diarrhea started 10 hours to 13 days after the fair.
- Diarrhea lasted from 1 to 16 days.
- Of the 284 people who drank apple cider in the afternoon, 154 developed cryptosporidiosis.

- Of the 292 people who did not drink apple cider in the afternoon, only 6 developed cryptosporidiosis.
- *Cryptosporidium* was found in the apple cider, on the cider press, and in a stool specimen from a calf on the farm that supplied the apples.

The apples were probably contaminated by calf feces on the ground before or during harvest. The apples were probably not washed properly before they were pressed to make cider, so *Cryptosporidium* spread throughout the batch of cider.

Investigators warn farmers that farm animals should not graze in areas where fruits and vegetables are grown. Food processors should throw out any fruits and vegetables that have feces on them, and should wash fruits and vegetables thoroughly before they process them.

Source: Millard, P. S., Gensheimer K. F., Addiss, D. G., Sosin, D. M., Beckett, G. A., Houck-Jankoski, A., and Hudson, A. An outbreak of cryptosporidiosis from fresh-pressed apple cider. *J. Am. Med. Assoc.* 272(20): 1592.